

TriSphere

Complete Feature Directory

Complete overview of dashboard capabilities across all user roles

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1. Student Dashboard (Gamified & Wellness-Focused)

Home Hub

- **Timetable & Study Streak:** Weekly visual calendar showing upcoming lectures and study schedules.
- **Daily Planner:** Integration of tasks, quizzes, and simulations due in the current week.
- **ASTRA Daily Check-in:** Real-time empathetic voice agent for mood check-ins with selfie-camera opt-in visual emotion tracking (Llama 4 Vision).

Learn & Simulations

- **Interactive Learn Pane:** Browse chapters, watch lectures, and download textbook materials.
- **Engagement-Locked Player:** Tracks tab-switching or video skips during lectures and flags warnings.
- **Simulation Lab:** Integrated PhET HTML5 Science simulations and GeoGebra Math tools inside the viewport.
- **Lab Notebook & Evidence:** Data recorder sheet for parameters and direct verification screenshot uploader.

Gamification & AI Assistance

- **Lernix AI Chatbot:** Secure daily-limit tracked AI companion in Firestore to prevent local storage bypass.
- **Milestone Badges:** Collectible avatar profile cards (e.g. Astronaut, Wizard, Superhero) based on study XP.
- **Rewards Store:** Exchange earned study XP for profile frames and student customization perks.
- **Class Leaderboard:** Real-time XP-based ranking across classroom peers.

Tools & Discussions

- **Student Tools:** PDF markup reader, scientific calculator, notes pad, and coordinate grapher.
- **Chat & Forums:** Private 1-on-1 messaging with faculty and classmates, plus public category forum boards.

2. Teacher Dashboard (Management & Grading Hub)

Overview & Material Builder

- **Resource Center:** Publish lessons, syllabus timelines, and upload textbook PDF resources targeted by class/grade.
- **Analytics Dashboard:** Real-time overview of active student counts, pending tasks, and meeting requests.

Assignment & Lab Creator

- **Interactive Builder:** Form to select, description, and assign PhET or GeoGebra virtual experiments.
- **Parameter Configurator:** Define variables that students must log during simulations (e.g. Voltage, Current).
- **Subject Lockdown:** Automatically constraints subject selection to the teacher's subject (Chemistry, Physics, Biology) with developer sandbox bypass.

Grading & Class Directory

- **Grading Center:** Interface to review textbook homework, assign grades, and submit feedback comments.
- **Simulation Lab Grading Overlay:** Displays student parameter values side-by-side with fullscreen lightbox previews of verification screenshots.
- **Student Directory:** Inspect student progress cards, quiz logs, and ASTRA mood tracking check-in histories.
- **Announcements Broadcaster:** Form to post class-wide notices and reminders.

3. Parent Dashboard (Safety & Academic Oversight)

Child Activity & Progress Tracking

- **Home Performance:** Instant overview of average quiz scores, streak status, active alerts, and study hours.
- **Engagement Charts:** Time tracking breakdowns showing duration spent on textbooks, quizzes, and Lernix AI.
- **Grade Visualizers:** Trend graphs charting child's academic scores over time.
- **Curriculum Syllabus Reports:** Track lesson completion rates and syllabus progress.

Wellness & Communication

- **ASTRA Mood Analysis:** Log history showing emotional check-in outcomes categorized into wellness ranges (positive, neutral, sensitive mood triggers).
- **Teacher Bookings:** Scheduler to request and book consultation conferences with school teachers.
- **School Broadcasts:** Direct access to school-wide and class announcements.

4. Admin Dashboard (Principal Hub & System Console)

User & System Administration

- **Global Dashboard:** Core system statistics showing total students, faculty, and guardian counts.
- **User Directories:** Searchable, editable registers for Students, Parents, and Teachers (including lesson plan access).
- **Enrollment Module:** Register new user accounts and assign default roles (Admins, Teachers, Students, Parents).

Academic Reports & Broadcasts

- **School Progress:** View grade distributions, average scores, and curriculum tracking charts.
- **Broadcast Console:** Broadcast administrative bulletins and emergency alerts to all roles.

Integrity Logs & Content Management

- **Video Integrity Audit:** Real-time logs detailing tab-switches, skipped videos, or lecture url violations during assigned video watch periods.
- **Knowledge Base Manager:** Control panel to edit school FAQs, custom guide pages, and AI-guided replies.

